

“Live from LA!”: Facilitating Personal and Political Storytelling through Immersive Youth Theater

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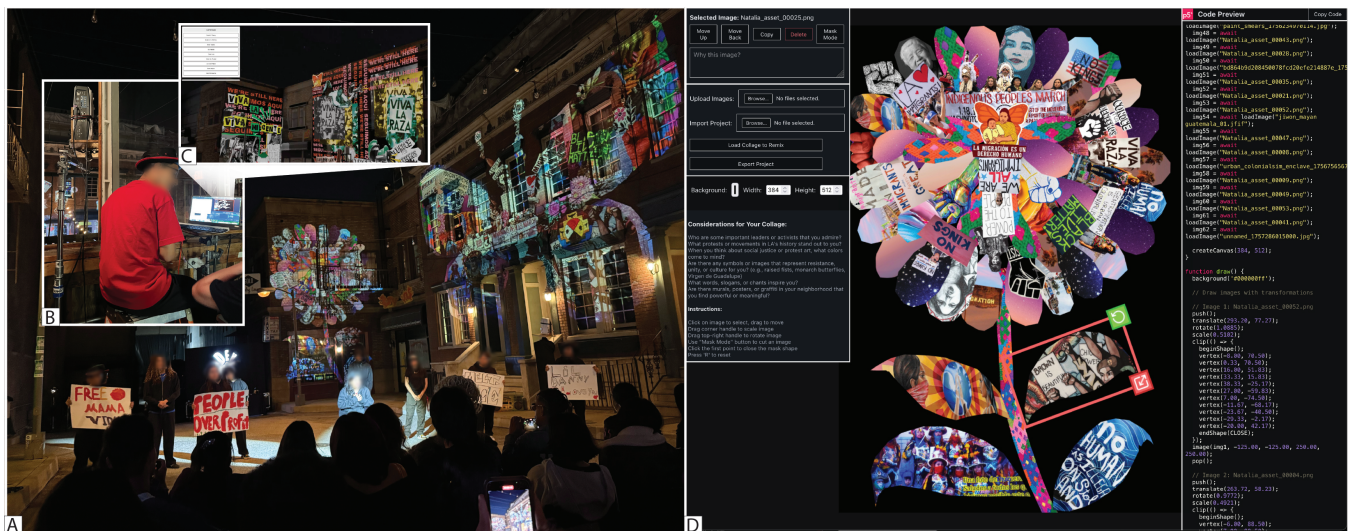


Figure 1: High school students perform *Live from LA!* in a studio backlot in LA, with projection-mapped visuals co-created with the youth ensemble (A). A student visits the control backstage (B), where we used a p5.js based software for controlling the projection mapping (C). The performance animations were based on collages created by students using a p5.js based tool. On performance day, a variant of the tool let audience members remix collages and images contributed by the ensemble (D).

Abstract

We present *Live from LA!*, an immersive theater production co-created with LA-based high school students and youth arts organizations, blending dance, poetry, theater, and Augmented Reality. The play unfolds against the backdrop of a community endangered by racial, economic, and political violence, foregrounding generational conflicts over how best to protect each other. To support youth-led personal and political storytelling, we co-created projection mapping animations using tools built with p5.js. Students used

these browser-based tools to manipulate visual parameters without requiring extensive prior coding experience. We discuss how this production was scaffolded across disciplines, locations, and age groups to support youth political storytelling through personal narratives. We highlight how project-specific p5.js tools supported youth authorship and politically grounded storytelling while navigating practical constraints and safety around sensitive narratives.

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1 Introduction

Augmented reality (AR) is an increasingly accessible medium for place-based cultural storytelling [MacIntyre et al. 2001; Shin and

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Woo 2023]. The technology can anchor stories to significant neighborhoods, institutions and contested sites—blending history with possible futures—while involving audiences in live, evolving narratives. AR projects in Los Angeles illustrate this civic potential across museums and public culture [LAC 2024; Lushetich and Fujihata 2020], interactive and live performance [Burke et al. 2022], interventions in public debates and activism [Jao and Monnier 2020; Museum 2025], alongside recent community responses to civil rights violations and political violence [Frattini 2026; Namkung 2025].

These interventions point to tensions in AR storytelling that is politically engaged, personally vulnerable, or tied to ongoing structural injustices and social conflict [Cárdenas Gasca et al. 2022]. Such political uses of AR become increasingly risky when led by youth sharing stories about their personal identities, lived experiences, and connections to physical locations. Responsible development of AR demands attention to whose narratives are represented, who bears risk, and what infrastructures govern visibility and control [Cárdenas Gasca et al. 2025].

We examine these tensions in a production-scale setting in a collaborative production with high school students that integrated dance, music, theater, and augmented reality in the creation of an immersive theater play: *Live from LA!* This production was a first-time collaboration between public non-profit youth arts organizations and an open-source creative coding foundation, with the objective of empowering young people to apply emerging and established creative arts media to tell stories that reflected their personal and political reality.

As part of the production process, we co-created projection mapping animations to ground the play's action in historical and cultural context. These effects were created using tools built with p5.js¹, an open-source library for creative coding designed to make programming accessible to artists and learners. Students used these browser-based tools to manipulate visual parameters in real-time, experimenting with color, movement, and composition without requiring extensive prior coding experience. This tool provided direct manipulation controls that map to core p5.js primitives and operations providing students with an initial low technical complexity to express their vision, while also creating on-ramps to engage with and learn from the underlying code.

We will present our experience scaffolding a collaboration across disciplines, locations, and age groups, engaging with open-source creative code tools, and building from personal narratives to critically engage with current political and social issues relevant to our young participants.

2 Live from LA!

Live from LA! is a novel play that involved four LA-based community arts non-profits, one international and decentralized open source organization, three academic collaborators from a US-based R1 research institution, and an LA-based media artist. The project engaged local high school students aged 13-19 who attended in-person sessions to produce and rehearse the play over the summer of 2025.

The story centers on a group of teens who rally a frightened community to save a beloved cultural center from gentrification and

political threats in an LA LatinX neighborhood. The story unfolds amid racial, economic, and political violence, and a generational divide over what it means to protect the community. Students developed the script, creating characters based on themselves and recent news events. The results are a collective narrative in which students also produced original soundtracks, spoken word, and poetry.

In collaboration with students and other facilitators, we co-developed projection-mapped visuals to ground the action in historical and cultural context, using political imagery to link present threats to longer legacies of struggle and resilience. Throughout the scriptwriting process, we introduced students to core concepts in projection mapping and creative coding. All the while, we developed custom tools that would enable the high schoolers to contribute to the final projected visual effects. As rehearsals progressed, we leveraged these tools and developed additional questionnaires to source visual references and compositions for the development of the projection mapping visuals (Sec 3). The resulting performance was staged twice: once for a community preview, and then for an invited live audience of roughly 200 at a film/TV studio backlot in Los Angeles.

3 Open-source Tools for Scaffolding Co-creation with Youth

We created two web-based systems to support youth-authored projection mapping for the play. First, we developed a novel collage tool (Fig. 2-A)—a visual interface that allowed users to arrange images directly on the screen. As photos were added and edited, corresponding JavaScript code was generated in real time, allowing users to recreate their collages using the p5.js editor or their own development environment. Students and collaborators used this tool to create protest collages, which we later remixed to create the projection mapping visuals for the final performance (Fig. 1-A).

We used the p5.js code generated by our collage tool as a teaching scaffold in sessions with one student, where we explored practical concepts embedded in the student's collages – loading assets, positioning, scaling, layering, and other basic functions (Fig. 2-B). Our goal was to ensure that this student not only understood how the tool and collages were made, but also how to use p5.js moving forward. The tool was implemented in TypeScript with a React UI and a p5.js canvas, and collages could be exported as zip files to preserve assets for remixing. We leveraged this functionality to allow audience members to remix collages and images contributed by the theater ensemble (Fig. 1-D).

Secondly, we built a cueing and control interface to synchronize multi-screen collages and animations with the live performance (Fig. 1-B and C). We defined cues as initial and final animation parameters exposed in custom shaders to create animated effects of the still collages. On site, we ran one browser per projector, routed outputs via Open Broadcaster Software (OBS)², and used local-network WebSockets to trigger scenes across all browsers.

¹<https://p5js.org/>

²<https://obsproject.com/>

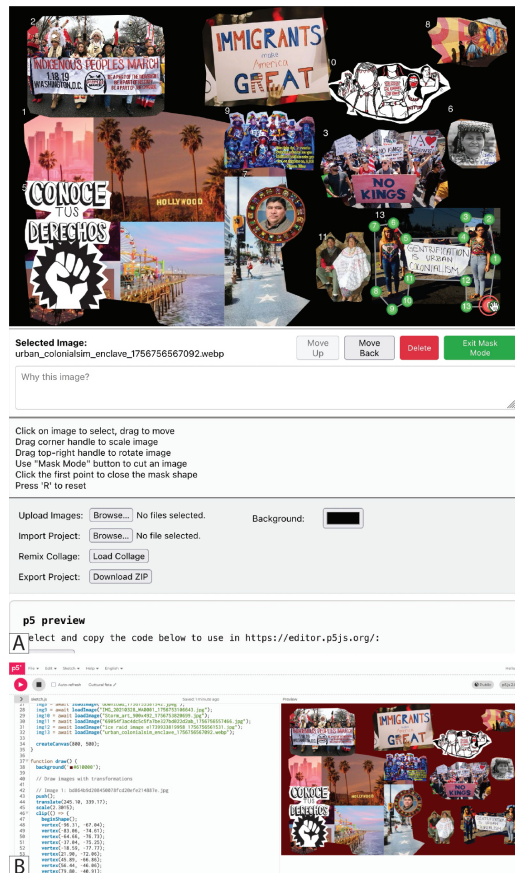


Figure 2: We supported the co-creation of visual effects with a p5.js-based direct-manipulation tools which allowed users to edit visual elements of a p5.js sketch, and export the p5.js code (A), we used this exported code to support creative coding lessons using p5.js (B).

4 Themes in the Production of Youth-led Immersive Narratives

During the play’s development, we identified four themes concerning the collaborative development of technologies for sensitive media narratives involving marginalized communities.

- (1) **Engaging in collaborative work between local and remote researchers, open-source organizations, and Arts NGOs.** This production introduced challenges and opportunities in the role of decentralized non-profit organizations in collaboration with local communities and organizations. Our role involved rapid prototyping through each phase of production and developing a shared visual and verbal language between partners with varying degrees of technical familiarity.
- (2) **Creative coding in multidisciplinary collaboration with youth engagement.** We scaffolded youth participation in technology development through project-specific authoring tools. This process included managing expectations for youth coding as one aspect of a broader multidisciplinary creative effort.

- (3) **Empowering youth in personally meaningful storytelling while preserving safety.** This project involved developing narratives that engage with current political issues from personal narratives and engaging with historical figures and themes that resonate with the young participants. In this process, tensions arose between providing young people with agency and authorship, without revealing sensitive information.
- (4) **Educational technology design for authentic creative contexts.** The co-creation process required surfacing pedagogical considerations when technology serves artistic vision rather than technology learning as the primary goal. We were able to position open-source tools as bridges between technical and non-technical collaborators by building tools that balance accessibility with creative expression.

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